

Partial Equilibrium

230333 Microeconomics 3 (CentER) – Part II
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Partial Equilibrium

- ▶ In the Robinson Crusoe economy we solved for general equilibrium in the special case of $L = 2$ and $I = J = 1$.
- ▶ Before considering the fully general case, we will study equilibria in only one good.
 - ▶ This is called *partial equilibrium*.
- ▶ Such an approach is reasonable when:
 1. The good makes up a small part of individuals' budgets, so wealth effects are negligible.
 2. Prices of all other goods in the economy are unaffected by changes in demand or supply of the good.

Jeremy Bentham (1748-1832)



- ▶ Born in London and educated at Oxford.
- ▶ Developed *felicific calculus*. Utility then had multiple dimensions, like intensity, duration, uncertainty, propinquity, fecundity, purity and extent. Nowadays we think of utility only having one dimension (like “intensity”).
- ▶ His body is preserved and left on display as an “auto-icon” at UCL.



William Stanley Jevons (1835-1882)

- ▶ Born in Liverpool and educated at UCL.
- ▶ Famous work is *A General Mathematical Theory of Political Economy* (1862).
- ▶ Reduced utility to two dimensions: intensity and time.
- ▶ Made the case that “economics, as a science concerned with quantities, is necessarily mathematical”. He was among the first to discuss marginal utility (or “final utility”).



Partial Equilibrium Setup: Consumers

- ▶ We consider the market for a single good ℓ and treat the other $L - 1$ goods as a composite commodity (e.g. money).
- ▶ We assume quasilinear utility over the composite commodity and good ℓ :

$$u_i(m_i, x_i) = m_i + \phi_i(x_i)$$

where m_i is i 's consumption of the composite good and x_i is i 's consumption of the good ℓ .

- ▶ With quasilinear utility, the wealth effects for x_i are zero.
- ▶ Assume ϕ_i is bounded above and $\phi'_i(x_i) > 0$ and $\phi''_i(x_i) < 0 \forall x_i \geq 0$.

Partial Equilibrium Setup: Consumers

- ▶ The price of good ℓ is p and the price of the composite is 1 (the numeraire).
- ▶ Assume that there is no initial endowment of good ℓ but $\omega_{mi} > 0 \forall i$ and $\sum_{i=1}^I \omega_{mi} = \bar{\omega}_m$.

Partial Equilibrium Setup: Firms

- ▶ A firm can use z_j units of the composite good to produce q_j units of good ℓ at cost $c_j(q_j)$
- ▶ $c'_j > 0$ and $c''_j > 0$ for all $q_j \geq 0$.
- ▶ Each firm therefore has the production set:

$$Y_j = \{(-z_j, q_j) : q_j \geq 0 \text{ and } z_j \geq c_j(q_j)\}$$

- ▶ Each consumer i owns a share $\theta_{ij} \in [0, 1]$ of each firm $j = 1, \dots, J$, entitling them to a θ_{ij} share of that firm's profits.

Consumer's Problem

- ▶ Each consumer i chooses $(m_i, x_i) \in \mathbb{R} \times \mathbb{R}_+$ to solve:

$$\max_{m_i \in \mathbb{R}, x_i \in \mathbb{R}_+} m_i + \phi_i(x_i)$$

$$\text{subject to } m_i + px_i \leq \omega_{mi} + \sum_{j=1}^J \theta_{ij} (pq_j - c_j(q_j))$$

- ▶ Note: if were to restrict $m_i \geq 0$, then demand for x_i may depend on ω_{mi} .
- ▶ $\sum_{j=1}^J \theta_{ij} (pq_j - c_j(q_j))$ is sum of profits consumer i receives from all J firms.

Consumer's Problem

- ▶ Utility is strictly increasing in both goods so the budget constraint will hold with equality.
- ▶ After substituting for m_i , the problem becomes:

$$\max_{x_i \in \mathbb{R}_+} \omega_{mi} + \sum_{j=1}^J \theta_{ij} (pq_j - c_j(q_j)) - px_i + \phi_i(x_i)$$

- ▶ We still have the $x_i \geq 0$ constraint.
- ▶ Omitting constant terms, the Lagrangian is $\mathcal{L}(x_i, \lambda) = \phi_i(x_i) - px_i + \lambda x_i$.
- ▶ The KT conditions are $\phi'_i(x_i) - p + \lambda = 0$ and $\lambda x_i = 0$ (with $\lambda \geq 0$).
- ▶ Therefore:

$$\begin{cases} \phi'_i(x_i) - p \leq 0 & \text{if } x_i = 0 \\ \phi'_i(x_i) - p = 0 & \text{if } x_i > 0 \end{cases}$$

Firm's Problem

- ▶ Given price p , firm j solves:

$$\max_{q_j \geq 0} p q_j - c_j(q_j)$$

- ▶ The first-order conditions for each firm are $p \leq c'_j(q_j)$, with equality if $q_j > 0$.

Equilibrium

- ▶ To find an equilibrium we need to find an allocation and price vector that satisfy:
 - ▶ Utility maximization.
 - ▶ Profit maximization.
 - ▶ Market clearing in both goods.
- ▶ The following Lemma will require us to only need to check for market clearing for good ℓ :

Lemma

If the allocation $(\mathbf{x}_1, \dots, \mathbf{x}_I, \mathbf{y}_1, \dots, \mathbf{y}_J)$ and price vector $\mathbf{p} \gg \mathbf{0}$ satisfy the market clearing condition for all goods $\ell \neq k$, and if every consumer's budget constraint is satisfied with equality, so that $\mathbf{p} \cdot \mathbf{x}_i = \mathbf{p} \cdot \boldsymbol{\omega}_i + \sum_{j=1}^J \theta_{ij} \mathbf{p} \cdot \mathbf{y}_j$ for all i , then the market for good k also clears.

Proof of Lemma

Add all consumers' budget constraints and rearrange:

$$\sum_{i=1}^I \sum_{\ell=1}^L p_{\ell} x_{\ell i} - \sum_{i=1}^I \sum_{\ell=1}^L p_{\ell} \omega_{\ell i} - \sum_{i=1}^I \sum_{j=1}^J \sum_{\ell=1}^L \theta_{ij} p_{\ell} y_{\ell j} = 0$$

$$\sum_{\ell=1}^L p_{\ell} \sum_{i=1}^I x_{\ell i} - \sum_{\ell=1}^L p_{\ell} \bar{\omega}_{\ell} - \sum_{\ell=1}^L \sum_{j=1}^J p_{\ell} y_{\ell j} = 0$$

$$\underbrace{\sum_{\ell \neq k} p_{\ell} \left(\sum_{i=1}^I x_{\ell i} - \bar{\omega}_{\ell} - \sum_{j=1}^J y_{\ell j} \right)}_{=0 \text{ by market clearing in all goods } \ell \neq k} = -p_k \underbrace{\left(\sum_{i=1}^I x_{ki} - \bar{\omega}_k - \sum_{j=1}^J y_{kj} \right)}_{\text{Must be zero since } p_k > 0 \Rightarrow \text{Market clearing in good } k}$$

Equilibrium

- ▶ Using the Lemma, the allocation $(x_1^*, \dots, x_I^*, q_1^*, \dots, q_J^*)$ and price p^* constitute a competitive equilibrium iff we have:
 1. $p^* \leq c'_j(q_j^*)$, with equality if $q_j^* > 0$, for all $j = 1, \dots, J$
 2. $\phi'_i(x_i^*) \leq p^*$, with equality if $x_i^* > 0$, for all $i = 1, \dots, I$
 3. $\sum_{i=1}^I x_i^* = \sum_{j=1}^J q_j^*$
- ▶ If $\max_i \{\phi'_i(0)\} > \min_j \{c'_j(0)\}$ we will have $\sum_{i=1}^I x_i^* > 0$ in equilibrium
 - ▶ We will see why shortly.
 - ▶ We will assume this is the case from now on.

Demand and Aggregate Demand

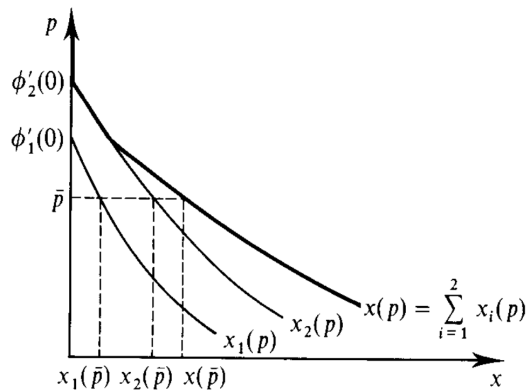
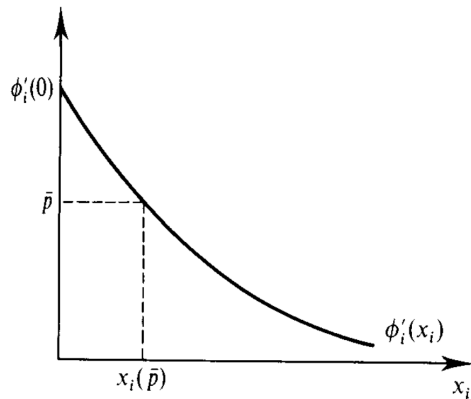
Individual Demand:

- ▶ Recall i 's FOC: $\phi'_i(x_i) \leq p$, with equality if $x_i > 0$.
- ▶ Since $\phi'_i > 0$ and $\phi''_i < 0$, ϕ'_i is positive and strictly decreasing.
- ▶ $\forall p > 0$, $\exists$ a unique x_i satisfying the FOC.
- ▶ This is $x_i(p)$, i 's demand function.
 - ▶ Doesn't depend on wealth (quasilinear utility).
- ▶ $x_i(p)$ is continuous and nonincreasing in p for all $p > 0$ and is strictly decreasing for $p < \phi'_i(0)$.

Aggregate Demand:

- ▶ Aggregate demand is then $x(p) = \sum_{i=1}^I x_i(p)$.
- ▶ $x(p) = 0$ for all $p > \max_i \{\phi'_i(0)\}$.
- ▶ $x(p)$ is continuous and nonincreasing for $p > 0$ and strictly decreasing for all $p < \max_i \{\phi'_i(0)\}$.

Demand and Aggregate Demand



Source: Mas-Colell, A., et al. (1995) *Microeconomic Theory*

Supply and Aggregate Supply

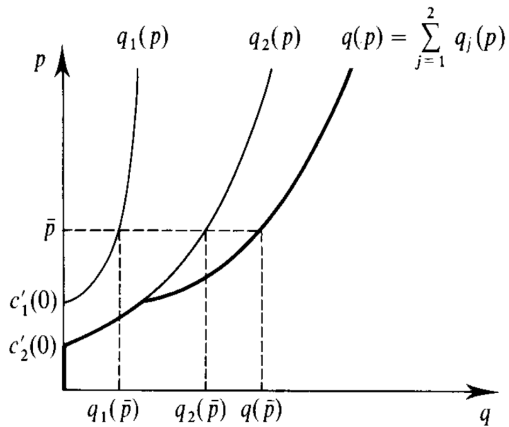
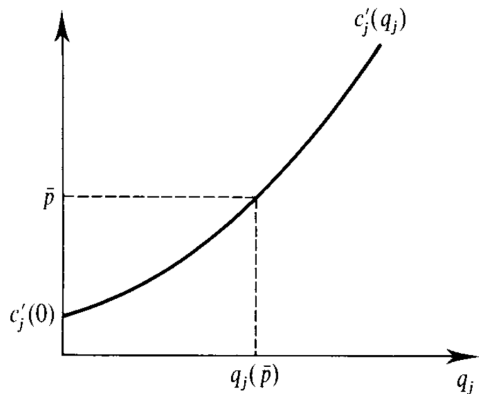
Individual Supply:

- ▶ Recall j 's FOC: $c'_j(q_j) \geq p$ with equality if $q_j > 0$.
- ▶ Since $c'_j > 0$ and $c''_j > 0$, c'_j is positive and strictly increasing.
- ▶ $\forall p > 0, \exists$ a unique q_j satisfying the FOC.
- ▶ This is $q_j(p)$, j 's supply function.
- ▶ $q_j(p)$ is continuous and nondecreasing at all $p > 0$ and is strictly increasing at any $p > c'_j(0)$.

Aggregate Supply:

- ▶ Aggregate supply is then $q(p) = \sum_{j=1}^J q_j(p)$.
- ▶ $q(p) = 0$ for all $p < \min_j \{c'_j(0)\}$.
- ▶ $q(p)$ is continuous and nondecreasing at all $p > 0$ and is strictly increasing at any $p > \min_j \{c'_j(0)\}$.

Supply and Aggregate Supply

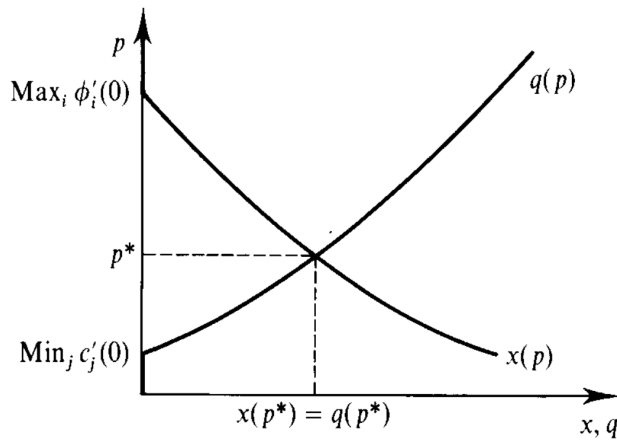


Source: Mas-Colell, A., et al. (1995) *Microeconomic Theory*

Equilibrium

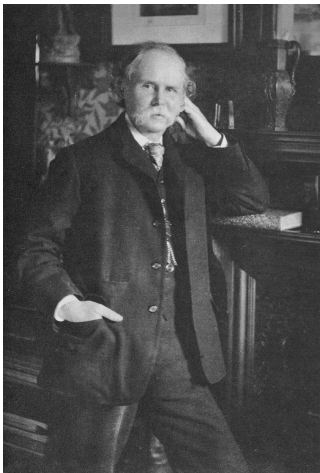
- ▶ Equilibrium occurs with a $p^\star$ satisfying $x(p^\star) - q(p^\star) = 0$.
- ▶ We assume $\max_i \{\phi'_i(0)\} > \min_j \{c'_j(0)\}$.
- ▶ There cannot be an equilibrium with either $p > \max_i \{\phi'_i(0)\}$ or $p < \min_j \{c'_j(0)\}$.
- ▶ At $p = \min_j \{c'_j(0)\}$, we have $x(p) > 0$ and $q(p) = 0$ so $x(p) - q(p) > 0$.
- ▶ At $p = \max_i \{\phi'_i(0)\}$, we have $x(p) = 0$ and $q(p) > 0$, so $x(p) - q(p) < 0$.
- ▶ Since $x(p) - q(p)$ is continuous and strictly decreasing, the existence of a *unique* equilibrium $p^\star$ is guaranteed.

Equilibrium

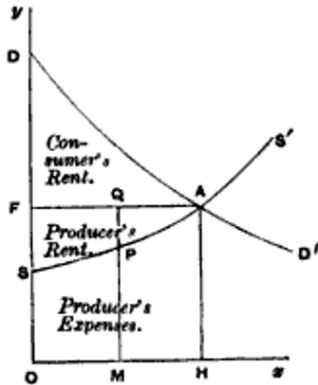


Source: Mas-Colell, A., et al. (1995) *Microeconomic Theory*

Alfred Marshall (1842-1924)



- ▶ Born in London and educated at Cambridge.
- ▶ One of the founders of neoclassical economics (the idea that prices are driven by supply and demand).
- ▶ His most famous work is the book *Principles of Economics* (1890), which popularized the use of diagrams in Economics. We usually put prices on the vertical axis, even though it is the independent variable in the demand function, because of the diagrams in this text.



Social Planner Problem

Assuming the social welfare function is the unweighted sum of utilities (Utilitarian social welfare function), the social planner has the following problem:

$$\begin{array}{l} \max \\ (x_1, \dots, x_I) \geq \mathbf{0} \\ (m_1, \dots, m_I) \\ (q_1, \dots, q_J) \geq \mathbf{0} \\ (z_1, \dots, z_J) \geq \mathbf{0} \end{array} \quad \sum_{i=1}^I [\phi_i(x_i) + m_i]$$

subject to:

1. $z_j \geq c_j(q_j)$ for all j (technology constraint for each firm).
2. $\sum_{i=1}^I m_i \leq \bar{\omega}_m - \sum_{j=1}^J z_j$ (feasibility for the composite commodity).
3. $\sum_{i=1}^I x_i \leq \sum_{j=1}^J q_j$ (feasibility for good ℓ).

Because utility is increasing in both goods, constraints 1-3 will bind. We can write:

$$\sum_{i=1}^I m_i = \bar{\omega}_m - \sum_{j=1}^J c_j(q_j)$$

Social Planner Problem

- ▶ Rewriting the problem with combined constraints:

$$\max_{\substack{(x_1, \dots, x_I) \geq \mathbf{0} \\ (q_1, \dots, q_J) \geq \mathbf{0}}} \sum_{i=1}^I \phi_i(x_i) + \bar{\omega}_m - \sum_{j=1}^J c_j(q_j)$$

$$\text{subject to } \sum_{i=1}^I x_i - \sum_{j=1}^J q_j = 0$$

- ▶ The first-order conditions are (with μ being the multiplier on the constraint):
 - ▶ $\mu \leq c'_j(q_j^\star)$ with equality if $q_j^\star > 0$, for $j = 1, \dots, J$.
 - ▶ $\phi'_i(x_i^\star) \leq \mu$ with equality if $x_i^\star > 0$, for $i = 1, \dots, I$.
 - ▶ $\sum_{i=1}^I x_i^\star = \sum_{j=1}^J q_j^\star$.
- ▶ These are precisely the equilibrium conditions as before with μ replacing $p^\star$.

The First Fundamental Theorem of Welfare Economics

- ▶ From this example, any competitive equilibrium must be Pareto optimal because it would satisfy the FOCs when $\mu = p^\star$.
- ▶ This is the first fundamental welfare theorem in the context of a two-good quasilinear model:

Theorem

If the price $p^\star$ and allocation $(x_1^\star, \dots, x_I^\star, q_1^\star, \dots, q_J^\star)$ constitute a competitive equilibrium, then this allocation is Pareto optimal.

The Second Fundamental Theorem of Welfare Economics

- ▶ We can redistribute endowments of the numeraire without affecting the production and demand for good ℓ .
- ▶ Therefore we can achieve any point on the utility possibility frontier as part of a competitive equilibrium by redistributing an appropriate amount of the numeraire to consumers.

Theorem

For any Pareto optimal levels of utility $(u_1^*, \dots, u_I^*)$, there are transfers of the numeraire commodity $(T_1, \dots, T_I)$ satisfying $\sum_i T_i = 0$, such that a competitive equilibrium reached from the endowments $(\omega_{m1} + T_1, \dots, \omega_{mI} + T_I)$ yields precisely the utilities $(u_1^*, \dots, u_I^*)$.

Later, we will prove this in the general equilibrium case.